

LOGICAL STRUCTURES

Lecture 5. Proof theory of predicate logic

Advanced Program in Computer Science

Spring, 2025

- ① Natural deduction rules
- ② Quantifier equivalences

Natural deduction rules

Proofs in the natural deduction calculus for predicate logic are similar to those for propositional logic in Chapter 1, except that we have new proof rules for dealing with the quantifiers and with the equality symbol.

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- ➊ Proof rules for equality
- ➋ Proof rules for universal quantification
- ➌ Proof rules for existential quantification

Proof rules for equality

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$$\frac{t_1 = t_2 \quad \phi[t_1/x]}{\phi[t_2/x]} =e$$

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Convention: Throughout this lecture, when we write a substitution in the form $\phi[t/x]$, we implicitly assume that t is free for x in ϕ .

Example 1.1

We obtain proof

1	$(x + 1) = (1 + x)$	premise
2	$(x + 1 > 1) \rightarrow (x + 1 > 0)$	premise
3	$(1 + x > 1) \rightarrow (1 + x > 0)$	=e 1, 2

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establishing the validity of the sequent

$$x + 1 = 1 + x, \quad (x + 1 > 1) \rightarrow (x + 1 > 0) \vdash (1 + x) > 1 \rightarrow (1 + x) > 0$$

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Example 1.2

Verify the validity of the sequent $t_1 = t_2 \vdash t_2 = t_1$

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Verify the validity of the sequent $t_1 = t_2 \vdash t_2 = t_1$

1	$t_1 = t_2$	premise
2	$t_1 = t_1$	=i
3	$t_2 = t_1$	=e 1, 2

where ϕ is $x = t_1$

Example 1.3

Verify the validity of the sequent $t_1 = t_2, t_2 = t_3 \vdash t_1 = t_3$

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	1	$t_2 = t_3$ premise
	2	$t_1 = t_2$ premise
	3	$t_1 = t_3$ =e 1, 2

where ϕ is $t_1 = x$

Proof rules for universal quantification

- *Rule for eliminating $\forall$* : if $\forall x\phi$ is true then you could replace the x in ϕ by any term t (remember the condition that t has to be free for x in ϕ), then $\phi[t/x]$ is true as well

$$\frac{\forall x \phi}{\phi[t/x]} \forall x e$$

Proof rules for universal quantification

- **Rule for introducing $\forall$:** if starting with a ‘fresh’ variable x_0 , you are able to prove some formula $\phi[x_0/x]$ then you can derive $\forall x \phi$

$$\frac{\boxed{\begin{array}{c} x_0 \\ \vdots \\ \phi[x_0/x] \end{array}}}{\forall x \phi} \forall x \text{ i}$$

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Verify the validity of the sequent $\vdash \forall x (P(x) \rightarrow P(x))$

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Example 1.4

Verify the validity of the sequent $\vdash \forall x (P(x) \rightarrow P(x))$

	x_0	
1	$P(x_0)$	assumption
2	$P(x_0) \rightarrow P(x_0)$	$\rightarrow_i$ 1–1
3	$\forall x (P(x) \rightarrow P(x))$	$\forall x i$ 1–2

Example 1.5

Give a proof for the sequent $\forall x(P(x) \rightarrow Q(x)), \forall x P(x) \vdash \forall x Q(x)$.

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1	$\forall x (P(x) \rightarrow Q(x))$	premise
2	$\forall x P(x)$	premise
3	$x_0 \quad P(x_0) \rightarrow Q(x_0)$	$\forall x \text{ e } 1$
4	$P(x_0)$	$\forall x \text{ e } 2$
5	$Q(x_0)$	$\rightarrow \text{e } 3, 4$
6	$\forall x Q(x)$	$\forall x \text{ i } 3-5$

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Verify the validity of the sequent

$$P(t), \forall x(P(x) \rightarrow \neg Q(x)) \vdash \neg Q(t)$$

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- | | | |
|---|---|---------------------|
| 1 | $P(t)$ | premise |
| 2 | $\forall x(P(x) \rightarrow \neg Q(x))$ | premise |
| 3 | $P(t) \rightarrow \neg Q(t)$ | $\forall$ e2 |
| 4 | $\neg Q(t)$ | $\rightarrow$ e3, 1 |

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Example 1.7

Is the following derivation correct?

- | | | |
|---|---|---------------------|
| 1 | $\forall x \exists y R(x, y)$ | premise |
| 2 | $\exists y R(y, y)$ | $\forall_e$ 1 |
| 3 | $\forall x \exists y R(x, y) \rightarrow \exists y R(y, y)$ | $\rightarrow_i$ 1–2 |

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Verify the validity of the sequent

$$P(t), \forall x(P(x) \rightarrow \neg Q(x)) \vdash \neg Q(t)$$

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| 1 | $P(t)$ | premise |
| 2 | $\forall x(P(x) \rightarrow \neg Q(x))$ | premise |
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It is wrong because y is not free for x in $\exists y R(x, y)$

Proof rules for existential quantification

- **Rule for introducing $\exists$:** (*exists-introduction*) we can deduce $\exists x\phi$ whenever we have $\phi[t/x]$ for some term t .

$$\frac{\phi[t/x]}{\exists x\phi} \exists x i.$$

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Example 1.8

Verify the validity of the sequent $\vdash R(a, a) \rightarrow \exists x R(x, x)$

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Example 1.8

Verify the validity of the sequent $\vdash R(a, a) \rightarrow \exists x R(x, x)$

1	$R(a, a)$	assumption
2	$\exists x R(x, x)$	$\exists x i$ 1
3	$R(a, a) \rightarrow \exists x R(x, x)$	$\rightarrow_i$ 1–2

Example 1.9

Prove the validity of the sequent $\forall xP(x) \vdash \exists xP(x)$

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- | | | |
|----|-----------------|-----------------|
| 1. | $\forall xP(x)$ | premise |
| 2. | $P(z)$ | $\forall x$ e 1 |
| 3. | $\exists xP(x)$ | $\exists x$ i 2 |

Proof rules for existential quantification

- *Rule for eliminating $\exists$* : if we know $\exists x\phi$ is true then ϕ is true for at least one value of x .

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- *Rule for eliminating $\exists$* : if we know $\exists x\phi$ is true then ϕ is true for at least one value of x . If assuming $\phi[x_0/x]$ allows us to prove some χ which does not mention x_0 , then this χ must be true whichever x_0 makes $\phi[x_0/x]$ true.

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- Rule for eliminating $\exists$:** if we know $\exists x\phi$ is true then ϕ is true for at least one value of x . If assuming $\phi[x_0/x]$ allows us to prove some χ which does not mention x_0 , then this χ must be true whichever x_0 makes $\phi[x_0/x]$ true.

$$\frac{\exists x \phi \quad \boxed{\begin{array}{c} x_0 \quad \phi[x_0/x] \\ \vdots \\ \chi \end{array}}}{\chi} \exists e$$

Example 1.10

Prove the validity of the sequent

$$\forall x(P(x) \rightarrow Q(x)), \exists x P(x) \vdash \exists x Q(x)$$

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Prove the validity of the sequent

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1	$\forall x (P(x) \rightarrow Q(x))$	premise
2	$\exists x P(x)$	premise
3	$x_0 \quad P(x_0)$	assumption
4	$P(x_0) \rightarrow Q(x_0)$	$\forall x$ e 1
5	$Q(x_0)$	$\rightarrow$ e 4, 3
6	$\exists x Q(x)$	$\exists x$ i 5
7	$\exists x Q(x)$	$\exists x$ e 2, 3–6

Note: x_0 can not escape the scope of the box which declares it. The following proof is **illegal**.

1	$\forall x (P(x) \rightarrow Q(x))$	premise
2	$\exists x P(x)$	premise
3	$x_0 \quad P(x_0)$	assumption
4	$P(x_0) \rightarrow Q(x_0)$	$\forall x \text{ e } 1$
5	$Q(x_0)$	$\rightarrow \text{e } 4, 3$
6	$Q(x_0)$	$\exists x \text{ e } 2, 3-5$
7	$\exists x Q(x)$	$\exists x \text{ i } 6$

Example 1.11

Give a proof for the following argument

'If all quakers are reformists and if there is a protestant who is also a quaker, then there must be a protestant who is also a reformist.'

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‘If all quakers are reformists and if there is a protestant who is also a quaker, then there must be a protestant who is also a reformist.’

$$\forall x(Q(x) \rightarrow R(x)), \exists x(P(x) \wedge Q(x)) \vdash \exists x(P(x) \wedge R(x))$$

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1	$\forall x (Q(x) \rightarrow R(x))$	premise
2	$\exists x (P(x) \wedge Q(x))$	premise
3	$x_0 \quad P(x_0) \wedge Q(x_0)$	assumption
4	$Q(x_0) \rightarrow R(x_0)$	$\forall x$ e 1
5	$Q(x_0)$	$\wedge$ e ₂ 3
6	$R(x_0)$	$\rightarrow$ e 4, 5
7	$P(x_0)$	$\wedge$ e ₁ 3
8	$P(x_0) \wedge R(x_0)$	$\wedge$ i 7, 6
9	$\exists x (P(x) \wedge R(x))$	$\exists x$ i 8
10	$\exists x (P(x) \wedge R(x))$	$\exists x$ e 2, 3–9

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1	$\exists x P(x)$	premise
2	$\forall x \forall y (P(x) \rightarrow Q(y))$	premise
3	y_0	
4	$x_0 \quad P(x_0)$	assumption
5	$\forall y (P(x_0) \rightarrow Q(y))$	$\forall x \text{ e } 2$
6	$P(x_0) \rightarrow Q(y_0)$	$\forall y \text{ e } 5$
7	$Q(y_0)$	$\rightarrow \text{e } 6, 4$
8	$Q(y_0)$	$\exists x \text{ e } 1, 4-7$
9	$\forall y Q(y)$	$\forall y \text{ i } 3-8$

Exercises 1

Give a proof for the following sequents

a. $R(a, b) \vdash \exists x \exists y R(x, y)$

b. $\forall x A(x) \vdash A(c) \wedge A(d)$

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b. $\forall x A(x) \vdash A(c) \wedge A(d)$

c. $\forall x (A(x) \wedge B(x)) \vdash \forall x B(x)$

d. $\exists x P(x) \rightarrow Q(c) \vdash \forall x (P(x) \rightarrow Q(c))$

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c. $\forall x (A(x) \wedge B(x)) \vdash \forall x B(x)$

d. $\exists x P(x) \rightarrow Q(c) \vdash \forall x (P(x) \rightarrow Q(c))$

e. $\forall x (P(x) \rightarrow \neg P(x)) \vdash \forall x \neg P(x)$

f. $\neg \exists x \neg A(x) \vdash \forall x A(x)$

Exercises 2

Give a proof for the following sequents

a. $\exists x (P(x) \wedge Q(x)) \vdash \exists x P(x)$

b. $\exists x R(x, x) \vdash \exists x \exists y R(x, y)$

Exercises 2

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a. $\exists x (P(x) \wedge Q(x)) \vdash \exists x P(x)$

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d. $\forall x (P(x) \rightarrow \neg P(x)) \vdash \neg \exists x P(x)$

Exercises 2

Give a proof for the following sequents

a. $\exists x (P(x) \wedge Q(x)) \vdash \exists x P(x)$

b. $\exists x R(x, x) \vdash \exists x \exists y R(x, y)$

c. $\forall x A(x) \vdash \neg \exists x \neg A(x)$

d. $\forall x (P(x) \rightarrow \neg P(x)) \vdash \neg \exists x P(x)$

e. $\exists x \forall y R(x, y) \vdash \forall y \exists x R(x, y)$

f. $\forall x (P(x) \vee Q(x)) \vdash \forall x P(x) \vee \exists x Q(x)$

Quantifier equivalences

Example 2.1

Translate sentence into predicate logic:

‘Not all bird can fly’

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- **Option 2:** $\exists x (B(x) \wedge \neg F(x))$

Quantifier equivalences

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Translate sentence into predicate logic:

‘Not all bird can fly’

- **Option 1:** $\neg \forall x (B(x) \rightarrow F(x))$
- **Option 2:** $\exists x (B(x) \wedge \neg F(x))$

Both are correct and they are *equivalence*. We write $\phi_1 \dashv\vdash \phi_2$ iff $\phi_1 \vdash \phi_2$ and $\phi_1 \dashv \phi_2$.

Theorem

Let ϕ and ψ be formulas of predicate logic. Then we have the following equivalences:

- ① (a) $\neg\forall x\phi \dashv\vdash \exists x\neg\phi$
- (b) $\neg\exists x\phi \dashv\vdash \forall x\neg\phi$

Theorem

Let ϕ and ψ be formulas of predicate logic. Then we have the following equivalences:

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 - (a) $\neg\forall x\phi \dashv\vdash \exists x\neg\phi$
 - (b) $\neg\exists x\phi \dashv\vdash \forall x\neg\phi$
- ② Assuming x is not free in ψ :
 - (a) $\forall x\phi \wedge \psi \dashv\vdash \forall x(\phi \wedge \psi)$ 
 - (b) $\forall x\phi \vee \psi \dashv\vdash \forall x(\phi \vee \psi)$
 - (c) $\exists x\phi \wedge \psi \dashv\vdash \exists x(\phi \wedge \psi)$
 - (d) $\exists x\phi \vee \psi \dashv\vdash \exists x(\phi \vee \psi)$
 - (e) $\forall x(\psi \rightarrow \phi) \dashv\vdash \psi \rightarrow \forall x\phi$
 - (f) $\exists x(\phi \rightarrow \psi) \dashv\vdash \forall x\phi \rightarrow \psi$
 - (g) $\forall x(\phi \rightarrow \psi) \dashv\vdash \exists x\phi \rightarrow \psi$
 - (h) $\exists x(\psi \rightarrow \phi) \dashv\vdash \psi \rightarrow \exists x\phi$.

Theorem

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$$(a) \forall x \phi \wedge \forall x \psi \dashv\vdash \forall x (\phi \wedge \psi)$$

$$(b) \exists x \phi \vee \exists x \psi \dashv\vdash \exists x (\phi \vee \psi)$$

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$$(a) \forall x \forall y \phi \dashv\vdash \forall y \forall x \phi$$

$$(b) \exists x \exists y \phi \dashv\vdash \exists y \exists x \phi$$